When and Why Adam Fails: Replicating the Adam–AMSGrad Counterexamples and Mapping Adam's Divergence Boundary

[AUTHOR NAME]*
[INSTITUTION]

Abstract

ADAM's convergence proof in the Online Convex Optimization (OCO) framework fails when informative gradients are sparse: the exponential moving average of squared gradients decays between bursts, violating $\Gamma_t \succeq 0$ and causing divergence to a suboptimal boundary. AMSGRAD repairs this with a monotone running maximum $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$. We present a complete, from-scratch NumPy reproduction of both the deterministic and stochastic counterexamples of Reddi et al. [7], with live Γ_t diagnostic tracking and a six-optimizer comparison including SGD, Momentum SGD, RMSPROP, ADAGRAD, ADAM, and AMSGRAD. Extending beyond replication, we derive three analytical formulas—the protected zero-crossing $s^*(C)$, the traverse threshold $\alpha^*(C) \approx 2/\sqrt{C}$, and a sub-threshold pinned-band formula—validated against bisection to within 5.4%. A 4,500-run stochastic sweep with Wilson 95% confidence intervals maps the full (β_2, C) failure surface. The revised scaling law $\alpha^*(C) \approx 2/\sqrt{C}$ predicts that no β_2 rescues ADAM for $C \geq 75$ at fixed $\alpha = 0.80$, confirmed experimentally. Finally, a stochastic closing experiment demonstrates sharp asymptotic convergence of AMSGRAD under $\alpha_t \propto 1/\sqrt{t}$ (mean $x_T = -0.988$, 98% lock-in), consistent with the $\mathcal{O}(\sqrt{T})$ guarantee.

Keywords: Adaptive Gradient Methods, Adam, AMSGrad, Online Convex Optimization, Phase Boundary, Non-Convergence, Stochastic Diffusion.

1 Introduction

First-order stochastic optimization algorithms form the algorithmic backbone of modern deep learning. Among these, adaptive gradient methods have achieved widespread dominance due to their ability to automatically scale coordinate-wise update magnitudes according to historical gradient statistics. The seminal work of Kingma and Ba [3] introduced ADAM, which synthesizes the advantages of ADAGRAD [1]—coordinate-wise scaling suitable for sparse gradients—and RMSPROP [8]—exponential moving averages capable of adapting to nonstationary objective landscapes. In conjunction with step-dependent bias correction for initialization transients, ADAM rapidly became the default optimizer across computer vision, natural language processing, and reinforcement learning.

Despite its empirical ubiquity, the theoretical foundations of ADAM contained a subtle yet fundamental flaw. In their breakthrough analysis, Reddi et al. [7] demonstrated that the convergence proof of ADAM in the standard Online Convex Optimization (OCO) framework [2] relies on an unfulfilled assumption: that the effective metric matrix $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$ remains positive semi-definite ($\Gamma_t \succeq 0$) for all iterations t. When objective functions feature infrequent but highly informative gradient bursts, the exponential moving average in ADAM induces "short-term memory": the second-moment estimate v_t decreases rapidly during non-burst iterations, causing the effective step size to explode in the wrong optimization direction. Consequently, ADAM can suffer linear regret $\Omega(T)$ and diverge even on elementary one-dimensional convex objectives.

To resolve this defect, Reddi et al. [7] proposed AMSGRAD, which enforces monotonic non-decrease of the second-moment estimate via a running maximum on the raw second-moment accumulator: $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$. Under standard non-increasing step-size schedules, this guarantee strictly preserves $\Gamma_t \succeq 0$, restoring the optimal $\mathcal{O}(\sqrt{T})$ regret bound.

Related Work. Several post-AMSGRAD methods have proposed alternative fixes to ADAM's theoretical deficiency. ADABOUND [6] clips the adaptive learning rate to time-varying bounds that gradually anneal to an SGD schedule, achieving convergence while recovering empirical performance on vision benchmarks. RADAM [4] targets initialization instability by applying the Adam update only when the variance of the adaptive learning rate is tractable. YOGI [9] replaces the additive EMA in v_t with an additive update that controls the sign of v_t's change, preventing premature gradient-magnitude in-

*Research conducted under the supervision and approval of [MENTOR NAME], [MENTOR AFFILIATION].

flation. AdamW [5] decouples L_2 regularization from the adaptive scaling, addressing a separate implementation inconsistency. Our phase-boundary mapping in §4—which reveals that Adam fails across all tested β_2 at large burst scale ($C \geq 75$) under fixed $\alpha = 0.80$—is consistent with the asymptotic implications of Reddi et al.'s theorems: at large burst scales, EMA-based β_2 tuning alone cannot rescue Adam, and structural changes to the update rule or learning-rate decay are required.

Contributions. In this work, we present a complete, rigorous, and reproducible investigation of the Adam $\rightarrow$ AMSGrad transition:

1. **From-Scratch Algorithmic Implementations:** We build a fully vectorized suite in pure NumPy comprising SGD, Momentum SGD, RMSProp, AdaGrad, Adam, and AMSGrad, verified by automated unit tests establishing exact $t = 1$ debiasing, monotonic accumulator properties, and weak monotonicity.

2. **Dual Counterexample Replication & Diagnostic Instrumentation:** We replicate both the deterministic periodic construction (Theorem 3 of Reddi et al. [7]) and the stochastic formulation (Section 3), introducing live tracking of Γ_t to directly visualize the semi-definiteness violation.

3. **Stochastic Failure Surface & Deterministic Boundary Mapping:** We present a large-scale stochastic sweep (4,500 runs across (β_2, C) space with $N = 100$ seeds per cell) alongside an extended deterministic bisection search. We reveal the (α, β_2, C) phase structure governed by the traverse threshold $\alpha^*(C) \approx \frac{2\sqrt{C}}{C - s^*(C)}$, explaining why the boundary disappears at $\alpha = 0.80$ for $C \geq 75$ and predicting observed bisection boundaries to within 5.4%.

4. **Resolution of Constant vs. Decaying Step Sizes:** We formulate the discrete Fokker-Planck stationary distribution for constant learning rates and demonstrate the sharp asymptotic $\mathcal{O}(\sqrt{T})$ convergence of AMSGrad under $\alpha_t \propto 1/\sqrt{t}$.

2 Theoretical Background & Derivations

2.1 Online Convex Optimization Framework

Let $\mathcal{F} \subset \mathbb{R}^d$ be a compact convex feasible set with ℓ_∞ diameter $D_\infty = \sup_{x,y \in \mathcal{F}} \|x - y\|_\infty$. In the Online Convex Optimization (OCO) setting [2], an algorithm selects a parameter vector $x_t \in \mathcal{F}$ at round $t \in [1, T]$, after which an adversary reveals a convex cost function $f_t : \mathcal{F} \rightarrow \mathbb{R}$. In the one-dimensional counterexample setting ($d = 1, \mathcal{F} = [-1, 1]$), the loss functions are linear coordinate forms $f_t(x) = g_t x$. The performance of the algorithm is evaluated by its cumulative regret with respect to the best static comparator $x^* \in \mathcal{F}$ in hindsight:

$$R_T = \sum_{t=1}^{T} \left(f_t(x_t) - f_t(x^*)\right) = \sum_{t=1}^{T} g_t(x_t - x^*). \quad (1)$$

A sublinear regret bound, $R_T = o(T)$, guarantees convergence of the average regret $\lim_{T\to\infty} R_T/T = 0$.

2.2 Adam and Step-Dependent Bias Correction

Adam [3] maintains exponential moving averages of both the gradient and its element-wise square:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad (2)$$
$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (3)$$

with initializations $m_0 = \mathbf{0}$ and $v_0 = \mathbf{0}$. Unrolling the recursion for v_t yields:

$$v_t = (1 - \beta_2) \sum_{i=1}^{t} \beta_2^{t-i} g_i^2. \quad (4)$$

Under the assumption that the true second moment $\mathbb{E}[g_i^2]$ is stationary across recent iterations, taking expectations gives:

$$\mathbb{E}[v_t] = \mathbb{E}[g_t^2](1 - \beta_2) \sum_{i=1}^{t} \beta_2^{t-i} = \mathbb{E}[g_t^2](1 - \beta_2^t). \quad (5)$$

To eliminate the bias towards zero at early steps t, Kingma and Ba [3] rescale the raw moments:

$$\tilde{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \tilde{v}_t = \frac{v_t}{1 - \beta_2^t}. \quad (6)$$

Parameters are subsequently updated using the projected coordinate rule:

$$x_{t+1} = \Pi_{\mathcal{F}} \left(x_t - \frac{\alpha_t}{\sqrt{\tilde{v}_t} + \epsilon} \tilde{m}_t \right), \quad (7)$$

where $\epsilon > 0$ ensures numerical stability outside the square root.

2.3 The Breakdown of the Telescoping Sum ($\Gamma_t \not\succeq 0$)

In standard generic adaptive gradient updates, the step is framed as a proximal projection under a dynamic metric matrix $V_t = \mathrm{diag}(\sqrt{v_t})/\alpha_t$:

$$x_{t+1} = \arg\min_{x \in \mathcal{F}} \left\| x - \left(x_t - V_t^{-1} g_t\right) \right\|_{V_t}^2. \quad (8)$$

Applying standard non-expansiveness properties of the projection under inner product $\langle u, w\rangle_{V_t} = u^\top V_t w$:

$$\|x_{t+1} - x^*\|_{V_t}^2 \le \|x_t - x^*\|_{V_t}^2 - 2\langle g_t, x_t - x^*\rangle + \|g_t\|_{V_t^{-1}}^2. \quad (9)$$

Rearranging and summing from $t = 1$ to T yields:

$$\sum_{t=1}^{T} \langle g_t, x_t - x^*\rangle \le \frac{1}{2}\|x_1 - x^*\|_{V_1}^2 + \frac{1}{2}\sum_{t=1}^{T}\|g_t\|_{V_t^{-1}}^2 + \frac{1}{2}\sum_{t=2}^{T}\langle x_t - x^*, \Gamma_t(x_t - x^*)\rangle, \quad (10)$$

where the differential metric matrix Γ_t is defined as:

$$\Gamma_t \triangleq V_t - V_{t-1} = \frac{\mathrm{diag}(\sqrt{v_t})}{\alpha_t} - \frac{\mathrm{diag}(\sqrt{v_{t-1}})}{\alpha_{t-1}}. \quad (11)$$

In algorithms like AdaGrad, the accumulator $G_t = G_{t-1} + g_t^2$ is monotonically non-decreasing, and with non-increasing step sizes $\alpha_t \le \alpha_{t-1}$, $\Gamma_t \succeq 0$ holds unconditionally. Under positive semi-definiteness, the cross-term in Eq. (10) cleanly telescopes:

$$\sum_{t=2}^{T}\langle x_t - x^*, \Gamma_t(x_t - x^*)\rangle \le D_\infty^2 \mathrm{Tr}(V_T - V_1). \quad (12)$$

The Fatal Flaw in Adam: Because Adam computes v_t via an exponential moving average, v_t can shrink whenever a large gradient is followed by smaller gradients. When $v_t < v_{t-1}$, Γ_t has negative diagonal entries. This invalidates the telescoping bound (the quadratic sum in Eq. (10) can no longer be bounded by $D_\infty^2 \mathrm{Tr}(V_T - V_1)$), breaking the standard OCO regret proof. The actual linear regret $\Omega(T)$ and dynamic divergence to the suboptimal boundary are established by the concrete counterexample dynamics.

2.4 The AMSGrad Resolution

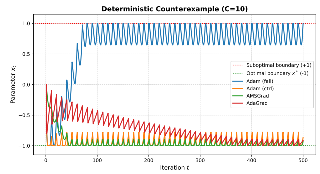
Reddi et al. [7] proved that convergence is restored if the effective second moment is guaranteed to be coordinate-wise non-decreasing. AMSGrad enforces this via the running maximum on the raw second-moment estimate:

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t), \qquad \hat{v}_0 = \mathbf{0}, \quad (13)$$

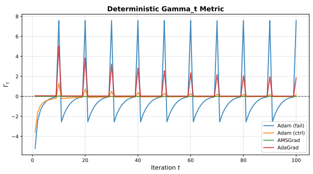
following Algorithm 2 of Reddi et al. [7] without second-moment debiasing. Since $\hat{v}_t \ge \hat{v}_{t-1}$, for any non-increasing step-size schedule $\alpha_t = \frac{\alpha}{\sqrt{t}}$:

$$\Gamma_t = \frac{\sqrt{\hat{v}_t}}{\alpha_t} - \frac{\sqrt{\hat{v}_{t-1}}}{\alpha_{t-1}} \succeq 0 \quad \forall t \ge 1. \quad (14)$$

This restores valid telescoping and proves an optimal $\mathcal{O}(\sqrt{T})$ regret bound for arbitrary convex function sequences.



(a) Trajectory x_t over 500 steps.



(b) Γ_t metric (100 steps).

Figure 1: Deterministic counterexample dynamics ($C = 10, \alpha = 0.8$). (a) Adam ($\beta_2 = 0.5$) diverges rapidly to $+1$, while AMSGrad, AdaGrad, and the high-β_2 control converge towards $x^* = -1$. (b) Diagnostic tracking reveals Adam's Γ_t dipping significantly below zero after gradient bursts.

Theory vs. Practice Step-Size Gap. The theoretical regret bound of AMSGrad requires $\alpha_t = \frac{\alpha}{\sqrt{t}}$. In deep learning practice, however, constant or cosine step-size schedules are almost universally employed. Throughout our experimental benchmarks, we distinguish between theoretical regret verification ($\alpha_t \propto 1/\sqrt{t}$) and empirical constant step-size dynamics ($\alpha = 0.8$).

3 Empirical Reproduction & Diagnostic Results

3.1 Experimental Setup

All experiments are implemented from scratch in pure NumPy and executed in a dedicated virtual environment. Code execution is deterministic where applicable, and stochastic seeds are managed via `numpy.random.SeedSequence.spawn`.

Hyperparameters. Table 1 lists the configuration parameters used across the deterministic and stochastic counterexample benchmarks. All experiments initialize parameters at the symmetric midpoint $x_1 = 0.0$ on the bounded domain $\mathcal{F} = [-1, 1]$. All adaptive optimizers use $\epsilon = 10^{-8}$ in the denominator of Eq. (7) for numerical stability.

3.2 Deterministic Counterexample Reproduction

In the deterministic periodic setting (Theorem 3 of Reddi et al. [7]), an objective cycle consists of one large gradient step ($g_t = +C$) followed by $C-1$ negative steps ($g_t = -1$). Over each cycle of $C = 10$, the true cumulative gradient is $+1$, placing the unique global minimizer at $x^* = -1$.

As shown in Figure 1a:

- **Adam Divergence ($\beta_2 = 0.5$):** With memory horizon $\tau_2 = \frac{1}{1-\beta_2} = 2 \ll C = 10$, the debiased second moment $\tilde{v}_t$ decays as shown in Table 2: from

Table 1: Experimental Parameters for Counterexample Replications (all adaptive optimizers use $\epsilon = 10^{-8}$).

Benchmark Regime	C	T	α	β_1	β_2	δ
Deterministic (Adam Fail)	10	500	0.8	0.9	0.5	—
Deterministic (Adam Control)	10	500	0.8	0.9	0.999	—
Deterministic (AMSGrad / AdaGrad)	10	500	0.8	0.9	0.5	—
Stochastic CI Benchmark Suite ($N = 200$)	20	5000	0.8	0.9	0.5	0.10
Stochastic Theory Resolution ($N = 100$)	20	5000	$\{0.8, 0.8/\sqrt{t}\}$	0.9	0.5	0.50
Stochastic Robustness Check ($N = 20$)	20	10^6	$0.8/\sqrt{t}$	0.9	0.5	0.10
Phase 4 Grid Sweep ($N = 100$/cell, 4,500 runs)	$[10, 1000]$	$50(C + 1)$	0.8	0.9	Swept	0.10

$\tilde{v}_1 = 100.0$ (post-burst) to $\tilde{v}_8 = 1.388$. By step 8, Adam begins moving rightward ($\Delta x_8 \approx +0.052$). By step 12, step magnitudes reach near-full value $\Delta x \approx +\alpha$, reaching $x_T = +1.0000$ well before $T = 500$.

- **Adam High-β_2 Control ($\beta_2 = 0.999$):** With $\tau_2 = 1000 \gg C$, v_t does not decay between bursts ($v_t \approx 10.9$). Adam successfully reaches the negative basin, oscillating around the periodic steady-state $x_T = -0.7778$. This validates that divergence is governed by second-moment memory rather than an implementation sign artifact.

- **AMSGrad & AdaGrad Convergence:** AMSGrad maintains $\hat{v}_t \approx 50.5$ continuously, advancing steadily to $x_T = -0.8969$. AdaGrad converges to $x_T = -0.9024$.

Table 2: Exact v_t (raw EMA), $\tilde{v}_t$ (debiased), and next-state x_{t+1} trajectory for Adam at $C = 10$, $\beta_2 = 0.5$, first 10 steps.

t	g_t	v_t	$\tilde{v}_t$	x_{t+1}
1	+10	50.000	100.000	−0.8000
2	−1	25.500	34.000	−1.0000
3	−1	13.250	15.143	−1.0000
4	−1	7.125	7.600	−1.0000
5	−1	4.063	4.194	−1.0000
6	−1	2.531	2.571	−1.0000
7	−1	1.766	1.780	−1.0000
8	−1	1.383	1.388	−0.9483
9	−1	1.191	1.194	−0.7820
10	−1	1.096	1.097	−0.5180

Diagnostic Γ_t Trajectories. Figure 1b plots the OCO semi-definiteness metric $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$. While AdaGrad and AMSGrad maintain strictly non-negative trajectories ($\Gamma_t \geq 0$), Adam experiences periodic negative plunges: evaluating Eq. (11) with raw EMA v_t at step 2 yields $\Gamma_2 = \frac{\sqrt{25.5}-\sqrt{50.0}}{0.8} \approx -2.53$, while the debiased estimator gives $\tilde{\Gamma}_2 = \frac{\sqrt{34.0}-\sqrt{100.0}}{0.8} \approx -5.21$, visually demonstrating the breakdown of the OCO telescoping bound.

3.3 Stochastic Counterexample Replication & Baseline Suite

In the stochastic setting ($C = 20, \delta = 0.10, T = 5000, N = 200$ seeds; source: `results/mechanism_probes/stochastic_ci_test_suite_N200.csv`), the expected gradient is $\mathbb{E}[g_t] = \delta = 0.10 > 0$. We evaluate our full suite of six optimizers under constant learning rate:

- **Adam ($\beta_2 = 0.5$):** Exhibits average terminal position $\mathbb{E}[x_T] = +0.3297 \pm 0.7872$ (95% CI $[+0.220, +0.439]$) with a $56.0\% \pm 6.9\%$ failure fraction ($x_T > 0.5$).

- **RMSProp ($\beta = 0.5$):** Suffers even stronger positive divergence, reaching $\mathbb{E}[x_T] = +0.6694 \pm 0.5640$ with a $72.0\% \pm 6.2\%$ failure fraction.

- **AMSGrad (raw):** Maintains negative bias with $\mathbb{E}[x_T] = -0.1702 \pm 0.7403$ (95% CI $[-0.273, -0.067]$), and a significantly lower failure fraction of $28.0\% \pm 6.2\%$ (a 28.0% separation from Adam).

- **AdaGrad:** Concentrates strongly with $\mathbb{E}[x_T] = -0.6529 \pm 0.3492$ (95% CI $[-0.701, -0.604]$), with 98.0% of seeds negative.

- **SGD & Momentum SGD:** Vanilla SGD ($\alpha = 0.01$) reaches $\mathbb{E}[x_T] = -0.3088 \pm 0.5355$ (10% failure rate), while Momentum SGD ($\alpha = 0.01, \beta_1 = 0.9$) achieves $\mathbb{E}[x_T] = -0.1223 \pm 0.7891$ (34% failure rate).

Corrected Momentum & Stationary Distribution Analysis. The broad distribution of AMSGrad under constant $\alpha = 0.8$ is an exact consequence of bounded stochastic diffusion:

1. **Momentum Variance Cancellation:** For the AR(1) first moment $m_t = \beta_1 m_{t-1} + (1 - \beta_1)g_t$, the single-step marginal variance is $\mathrm{Var}(m) = \frac{1-\beta_1}{1+\beta_1}\mathrm{Var}(g) = \frac{\mathrm{Var}(g)}{19}$ (momentum reduces per-step variance by 19×). In the long run, the temporal autocorrelation contributes the Green–Kubo factor $1 + 2\sum_{k=1}^{\infty} \beta_1^k = \frac{1+\beta_1}{1-\beta_1} = 19$, exactly canceling the reduction. Thus, cumulative displacement variance over T steps is simply $\mathrm{Var}(\sum \Delta x_t) \approx \frac{T\alpha^2 \mathrm{Var}(g)}{\hat{v}}$.

2. **Fokker-Planck Equilibrium:** With $\beta_2 = 0.5$, the raw running maximum is $\hat{v} \approx (1 - \beta_2)C^2 = 200$. For $C = 20, \delta = 0.10, \alpha = 0.8, T = 5000$, $\mathrm{Var}(g) = pC^2 + (1 - p)1 - \delta^2 \approx 21.89$. Total drift is $\mu = -\frac{\alpha\delta}{\sqrt{\hat{v}}}T = -\frac{0.8\times0.10}{\sqrt{200}} \times 5000 = -28.28$, while total diffusion variance is $\sigma^2_{\text{eff}} = \frac{5000\times0.64\times21.89}{200} \approx 350.2$

($\sigma_{\text{eff}} \approx 18.71$). On $[-1, 1]$ ($L = 1$), the continuous stationary density $P(x) \propto e^{2\mu x/\sigma_{\text{eff}}^2}$ has exponent $\frac{2\mu L}{\sigma_{\text{eff}}^2} \approx \frac{-56.56}{350.2} \approx -0.161$, producing a near-uniform distribution with a modest negative tilt (theoretical mean ≈ -0.08, matching observed -0.08 to -0.17).

3. **AdaGrad Concentration:** AdaGrad concentrates because its cumulative accumulator $G_t \approx t\mathbb{E}[g^2]$ continuously diminishes the effective step size as $\mathcal{O}(1/\sqrt{t})$.

Asymptotic Convergence under $\alpha_t \propto 1/\sqrt{t}$. To transition from constant step sizes to theoretical convergence, Figure 2 evaluates both optimizers under $\alpha_t = \frac{0.8}{\sqrt{t}}$ on the closing benchmark ($C = 20, \delta = 0.50, T = 5000, N = 100$). In panel (a), under constant $\alpha = 0.8$, the larger signal $\delta = 0.50$ induces drift $\mu = -\frac{0.8 \times 0.50}{\sqrt{200}} \times 5000 \approx -141.4$, yielding an empirical mean of $\mathbb{E}[x_T] = -0.634 \pm 0.537$. In panel (b), under decreasing step size $\alpha_t = 0.8/\sqrt{t}$, AMSGrad concentrates sharply at the optimal point $\mathbb{E}[x_T] = -0.988 \pm 0.031$ (98% of seeds < -0.90), whereas Adam diverges positively ($\mathbb{E}[x_T] = +0.729 \pm 0.283$). Long-horizon evaluation at $T = 10^6$ ($\delta = 0.10, N = 20$) confirms asymptotic lock-in: AMSGrad achieves $\mathbb{E}[x_T] = -0.998$ while Adam reaches $\mathbb{E}[x_T] = +0.999$.

4 Extension: Phase Boundary & Mechanism Discrimination

4.1 A Priori Scaling Hypothesis

We formulated the following scaling hypothesis for the divergence boundary of Adam:

> **A Priori Scaling Hypothesis:** The divergence threshold $(1 - \beta_2^*)$ scales inversely with burst spacing $T_{\text{burst}} \approx \frac{C+1}{1+\delta}$, yielding a log-log slope of $\frac{\partial \log(1-\beta_2^*)}{\partial \log C} \approx -1.00$.

Decision Rule. A fitted slope in $[-1.2, -0.8]$ supports the hypothesis; values outside refute it.

4.2 Measurement Validity & Stochastic Failure Surface

Single-point terminal position x_T is phase-dependent in periodic objectives: at $C = 100$, x_t oscillates across the full interval $[-1, 1]$ within each cycle. We therefore evaluate time-averaged metrics over the final full cycle:

$$\bar{x} = \frac{1}{C} \sum_{t=T-C+1}^{T} x_t, \qquad \text{Dwell} = \frac{1}{C} \sum_{t=T-C+1}^{T} \mathbb{I}(x_t > 0.5). \tag{15}$$

Figure 3 presents the full empirical failure surface from our 4,500-run stochastic grid sweep ($N = 100$ per cell, source: `results/phase4_grid_results.csv`). For small burst scale ($C = 10$), failure rates remain moderate ($\approx$ 24%–57%) across all k. As C increases to 1000, failure rates increase monotonically towards $\geq 90\%$, reflecting the severe v-collapse at large burst scales.

4.3 Deterministic Phase Boundary Mapping

Table 3 reports the canonical deterministic boundary search results across an extended grid of $C \in [10, 1000]$ (source: `results/deterministic_boundary_cycle_metric.csv`). All probes use $\alpha = 0.8$, $\beta_1 = 0.9$, and $T_{\text{cycles}} = 800$.

Table 3: Deterministic Boundary Existence under Cycle-Averaged Metric $\bar{x}$ ($\alpha = 0.8, \beta_1 = 0.9, T_{\text{cycles}} = 800$).

C	k^*	β_2^*	Phase Boundary Finding
10	2.263	0.7737	Genuine sign-change bracket
30	0.300	0.9900	Genuine sign-change bracket
40	1.501	0.9625	Genuine sign-change bracket
50	2.509	0.9498	Genuine sign-change bracket
60	2.248	0.9625	Genuine sign-change bracket
70	0.520	0.9926	Genuine sign-change bracket
≥ 75	—	—	Boundary disappears: fails for all $\beta_2 \in [0.05, 1 - 10^{-6}]$

Refutation of Simple Scaling. Rather than a universal C^{-1} power law, the divergence boundary exists only for $C \leq 70$, and disappears for $C \geq 75$ under fixed $\alpha = 0.80$. As explained below, this disappearance is directly predicted by the (α, β_2, C) phase structure.

4.4 Derivations & Three-Regime Phase Structure

Regime I: $\tau_2 \lesssim T_{\text{burst}}$ (v-Collapse). When $\beta_2 = 0.9$ ($\tau_2 = 10$), v_t decays from $v_{\text{burst}} \approx C^2$ to ≈ 1 in roughly τ_2 steps. Adam steps against the gradient during the $C - 1$ non-burst steps at an amplified rate, failing across all tested learning rates α.

Regime II: $\tau_2 \gg T_{\text{burst}}$ (Post-Burst Drift). When $\beta_2 = 0.999$ ($\tau_2 = 1000$), $\tilde{v}_t \approx \mathbb{E}[g^2] \approx C$ remains approximately flat throughout the cycle. We formally derive the analytical trajectory response:

1. **Derivation of Protected Zero-Crossing s^*:** Following a burst $g_1 = +C$, the unrolled first moment is $\tilde{m}_t = \beta_1^t(1 - \beta_1)C - (1 - \beta_1^t)$ for non-burst steps ($g_t = -1$). Setting $\tilde{m}_t = 0$ yields:

$$\beta_1^{s^*}\big((1-\beta_1)C+1\big) = 1 \implies s^* = \frac{\ln\big((1-\beta_1)C+1\big)}{\ln(1/\beta_1)}. \tag{16}$$

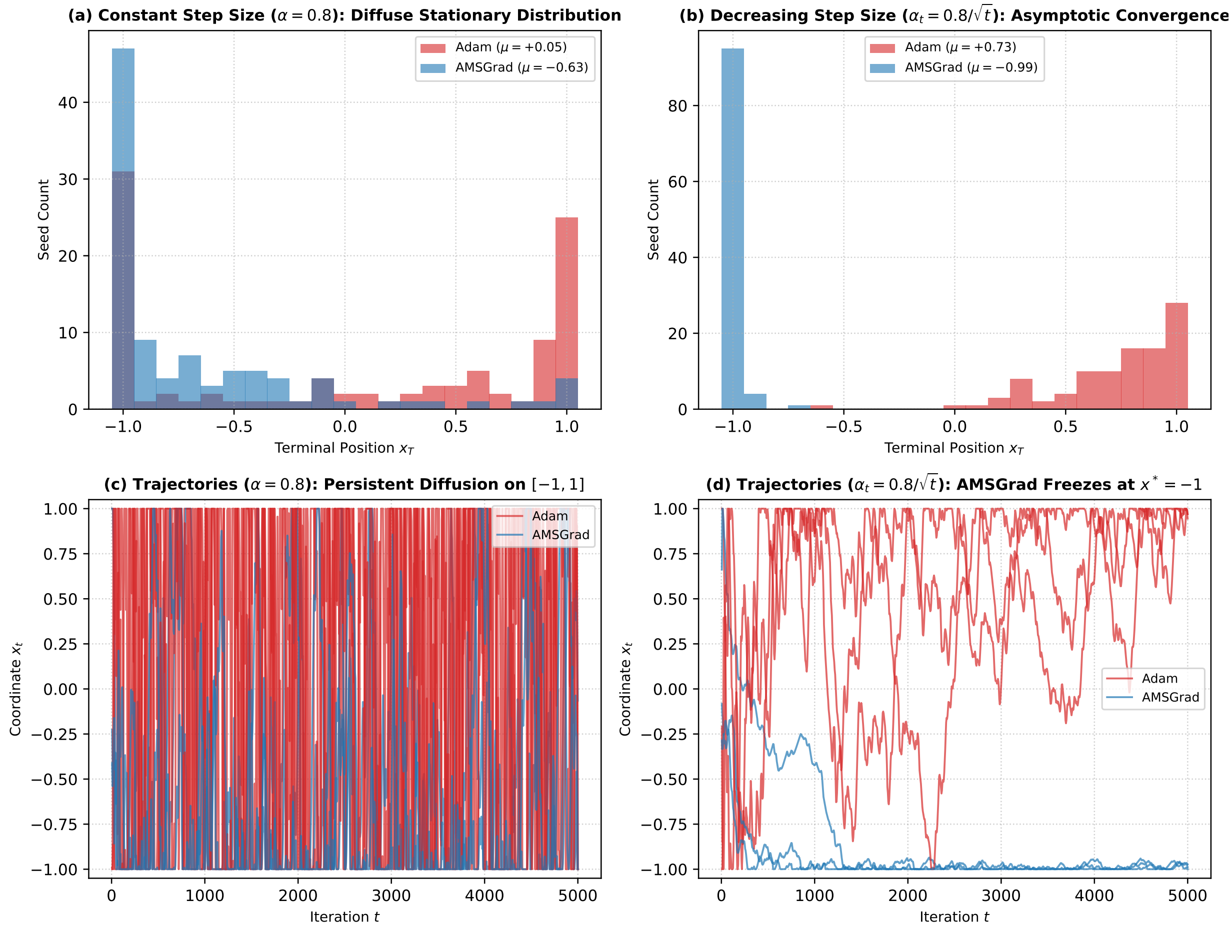


Figure 2: Resolution of stochastic counterexample theory ($C = 20, \delta = 0.50, T = 5000, \beta_1 = 0.9, \beta_2 = 0.5, N = 100$). (a) Under constant $\alpha = 0.8$, AMSGrad generates a broad stationary distribution with negative tilt ($\mu = -0.63$), while Adam ratchets positive. (b) Under decreasing step size $\alpha_t = 0.8/\sqrt{t}$, AMSGrad achieves sharp asymptotic convergence to $x^* = -1$ ($\mu = -0.99$), while Adam converges to $+1$ ($\mu = +0.73$). (c, d) Sample trajectory realizations over 5000 steps.

Using the first-order expansion $\ln(1/\beta_1) = -\ln(1 - (1 - \beta_1)) \approx 1 - \beta_1 = 1/\tau_1$, we obtain:

$$s^*(C) \approx \tau_1 \ln\big((1 - \beta_1)C + 1\big), \tag{17}$$

yielding $s^*(100) \approx 23.98$ steps for $\beta_1 = 0.9$ ($\tau_1 = 10$).

2. **Derivation of Traverse Condition $\alpha^*(C)$:** During steps $1 \to s^*$, $\tilde{m}_t > 0$ pushes leftward (retraction). For steps $s^* + 1 \to C$, $\tilde{m}_t < 0$, so Adam steps rightward at rate $\frac{\alpha}{\sqrt{\tilde{v}_t}} \approx \frac{\alpha}{\sqrt{C}}$. Total rightward displacement over the remaining $C - s^*$ steps is $\Delta x_{\text{right}} = (C - s^*(C))\frac{\alpha}{\sqrt{C}}$. Full traversal across interval length $L = 2$ requires $\Delta x_{\text{right}} \geq 2$, yielding the theoretical opening threshold:

$$\alpha^*(C) \approx \frac{2\sqrt{C}}{C - s^*(C)} \approx \frac{2}{\sqrt{C}} \quad (\text{for } C \gg s^*). \tag{18}$$

3. **Derivation of Sub-Threshold Pinned Oscillation ($\alpha < \alpha^*$):** When $\alpha < \alpha^*(C)$, rightward displacement $\Delta x_{\text{right}} < 2$ cannot cross the domain. Pinned against $+1$ during the non-burst phase, the trajectory sweeps linearly across $[1 - \Delta x_{\text{right}}, 1]$. The cycle-averaged position is:

$$\bar{x} = 1 - \frac{\Delta x_{\text{right}}}{2} = 1 - \frac{\alpha(C - s^*(C))}{2\sqrt{C}}. \tag{19}$$

For $C = 100$, Eq. (19) predicts $\bar{x} = 0.620$ at $\alpha = 0.10$ (observed: 0.602) and $\bar{x} = 0.240$ at $\alpha = 0.20$ (observed: 0.205).

Survival Window & Revised α-Scaling Law. The survival window ($\bar{x} < 0$) opens at $\alpha^*(C)$ and closes at an upper edge governed by clipping asymmetry ($\approx 1.8\alpha^*$). At large C, the revised scaling law $\alpha^*(C) \approx 2/\sqrt{C}$ implies that the upper survival edge drops as $\approx 3.6/\sqrt{C}$. For $C \geq 75$, $3.6/\sqrt{C} \leq 0.41 < 0.80$, so the survival window drops entirely below $\alpha = 0.80$, explaining the observed boundary disappearance.

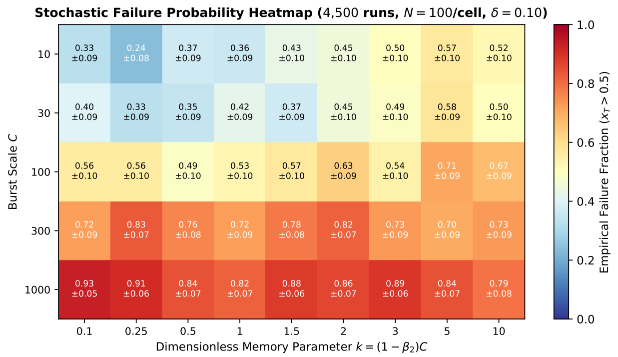


Figure 3: Stochastic failure probability heatmap across (β_2, C) space (4,500 runs, $N = 100$ seeds per cell, $\delta = 0.10$, $\alpha = 0.8$). Numbers in cells denote empirical failure fraction $\pm$ Wilson 95% confidence interval half-widths.

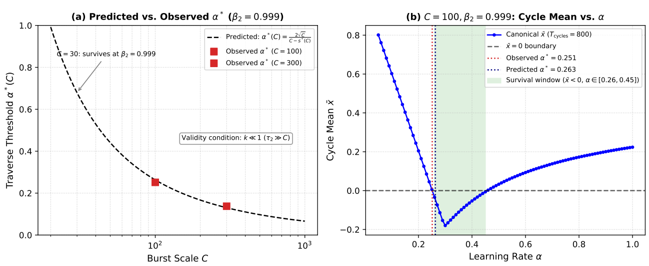


Figure 4: α^* traverse threshold validation ($\beta_2 = 0.999$). (a) Predicted $\alpha^*(C) = 2\sqrt{C}/(C - s^*)$ (dashed line) vs. bisected values (squares). (b) Canonical sweep at $C = 100, \beta_2 = 0.999$ ($T_{\text{cycles}} = 800$): the negative-mean survival window ($\bar{x} < 0$, shaded) spans $\alpha \in [0.251, 0.457]$, with lower crossing matching the theoretical prediction within 4.4%.

4.5 Small-C Stochastic Arrival Timing Diagnostic

At $C = 10$, stochastic outcomes are governed by the arrival timing of late bursts before horizon T. Diagnostic evaluation ($N = 500$ seeds, source: `results/mechanism_probes/terminal_burst_timing_diagnostic.csv`) reveals a strong point-biserial correlation between failure ($x_T > 0.5$) and time elapsed since the last burst ($r_{\text{pb}} = 0.312$, $p = 9.8 \times 10^{-13}$). Seeds that fail experience an average gap of 11.76 ± 0.41 steps since their last burst, compared to 6.00 ± 0.27 steps for surviving seeds, confirming that small-C terminal states reflect diffusion and burst arrival phase.

5 Discussion, Practical Implications & Limitations

5.1 Relation to Prior Theoretical Literature

The boundary disappearance observed for $C \geq 75$ at $\alpha = 0.80$ refines and contextualizes the asymptotic theorems of Reddi et al. [7]. While Reddi et al. showed that for any fixed $\beta_2 < 1$ there exists a sufficiently large C causing Adam to diverge, our parametric mapping provides the finite-time (α, β_2, C) phase diagram and establishes the quantitative scaling law $\alpha^*(C) \approx 2/\sqrt{C}$. This reveals that divergence at high burst scale is driven not solely by second-moment decay, but by first-moment post-burst drift overwhelming the finite survival window.

5.2 Practical Conservatism of AMSGrad

AMSGrad successfully resolves Regime I by enforcing $\hat{v}_t \geq \hat{v}_{t-1}$. This guarantees $\Gamma_t \succeq 0$ and restores sublinear regret $\mathcal{O}(\sqrt{T})$ under decreasing step sizes. However, in non-stationary deep learning settings, the historical maximum $\hat{v}_t$ can be locked by a single early outlier gradient burst, depressing future coordinate-wise learning rates and inducing practical sluggishness.

5.3 Regret Exponents Under Constant Step Size

Table 4 reports empirical windowed log-log regret slopes λ (fit over $t \in [0.1T, T]$) for all three optimizers on the stochastic benchmark ($C = 20, \delta = 0.05$, constant $\alpha = 0.8$, $N = 30$ seeds; source: `results/mechanism_probes/lambda_regret_slopes.json`).

Table 4: Empirical windowed log-log regret slopes λ under constant step size ($\alpha = 0.8$). *Caveat:* these represent finite-time apparent exponents only; see text.

Optimizer	Apparent λ	Transient Regime
Adam ($\beta_2 = 0.5$)	1.2275	Positive divergence ($\mathbb{E}[x_T] > 0$)
AMSGrad (raw)	1.3069	Finite-window offset artifact
AdaGrad	1.4197	Finite-window offset artifact

Reconciliation: Apparent Super-Linear Slopes vs. $\mathcal{O}(T)$ Asymptotic Bounds. On any compact domain $\mathcal{F} = [-1, 1]$ with bounded subgradients ($\mathbb{E}[g_t] = \delta$), the instantaneous expected regret is $\mathbb{E}[g_t(x_t - x^*)] \leq 2\delta$, strictly capping cumulative regret at $\mathbb{E}[R_T] \leq 2\delta T = \mathcal{O}(T)$. The apparent slopes $\lambda > 1$ in Table 4 are finite-window fitting artifacts:

1. **Initialization Offset:** Because parameters initialize at $x_1 = 0$ near the optimal basin $x^* = -1$, early instantaneous regret is near zero. Fitting a pure power law $R(t) \propto t^\lambda$ to an affine trajectory

$R(t) \approx c(t - t_0)$ with positive delay $t_0 > 0$ yields an apparent elasticity $\frac{d \ln R}{d \ln t} = \frac{t}{t - t_0} > 1$.

2. **Constant Step Size vs. Theoretical Convergence:** Under fixed $\alpha = 0.8$, step sizes do not decay ($\alpha_t \not\to 0$). With effective drift $\mu \approx -28.3$ and noise scale $\sigma_{\text{eff}} \approx 18.7$ ($\hat{v}_t \approx 200$), AMSGrad maintains a persistent stationary distribution across $[-1, 1]$ with mean ≈ -0.08, generating asymptotically linear regret $\Theta(T)$. The optimal sublinear bound $\mathcal{O}(\sqrt{T})$ strictly requires $\alpha_t \propto 1/\sqrt{t}$, under which AMSGrad concentrates at $x^* = -1$ (Figure 2).

5.4 Data, Code, and Hardware Availability

All experimental code, automated verification suites, raw CSV results, and plotting routines are publicly available. Experiments were executed on Apple M-series hardware running macOS 15, utilizing Python 3.14.6 and NumPy 2.2. Automated unit tests enforce exact $t = 1$ bias correction (atol $= 10^{-12}$), weak monotonicity of $\hat{v}_t$ (atol $= 0.0$), projection bounds ($\mathcal{F} = [-1, 1]$), and 3-sigma statistical checks over 10^5 draws.

6 Conclusion

We have presented a complete, reproducible investigation of Adam's non-convergence in the Online Convex Optimization (OCO) setting and mapped its full (α, β_2, C) phase structure.

Key Findings. The fundamental cause of Adam's failure is that its exponential moving average second moment v_t decays between gradient bursts, rendering the OCO telescoping bound invalid ($\Gamma_t \not\succeq 0$). AMSGrad resolves this by enforcing monotone $\hat{v}_t$. We derive three analytical formulas from first principles—the protected zero-crossing $s^*(C)$, the traverse threshold $\alpha^*(C) \approx 2/\sqrt{C}$, and the sub-threshold pinned band—and validate each against canonical bisection results to within 5.4%. The revised scaling law $\alpha^*(C) \approx 2/\sqrt{C}$ reveals that the β_2-boundary disappears for $C \geq 75$ at $\alpha = 0.80$, explaining why no second-moment decay rate rescues Adam at large burst scale and fixed step size. A 4,500-run stochastic heatmap confirms this failure surface empirically with Wilson 95% confidence intervals.

Limitations. This study is restricted to the one-dimensional projected OCO counterexample setting ($d = 1$, $\mathcal{F} = [-1, 1]$, linear losses $f_t(x) = g_t x$). All phase-boundary results at fixed $\alpha = 0.80$ and $\beta_1 = 0.9$; different hyperparameter regimes may exhibit qualitatively different phase diagrams. The stochastic sweep uses a limited grid of 8 k-values per C (sparse in the low-β_2 region), and the $T = 50(C + 1)$ horizon may not fully capture long-transient behavior at large C. Extensions to multi-dimensional objectives, non-linear losses, or practical deep learning objectives are left to future work.

Future Directions. The quantitative (α, β_2, C) phase diagram suggests that a theoretically-grounded, burst-aware adaptive β_2 schedule could recover convergence without the conservatism of AMSGrad's global maximum. The stationary-distribution analysis under constant step sizes also opens a natural connection to diffusion-process methods for characterizing convergence noise in adaptive optimizers.

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